



HardieFlex[®]
WET AREA WALLS

Installation Guide

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A better way to build.

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Additional installation information, warranties and warnings are available at www.jameshardie.com.ph

PHILIPPINES | JULY 2011

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WE VALUE YOUR FEEDBACK

To continue with the development of our products and systems, we value your input. Please send any suggestions, including your name, contact details, and relevant sketches to:

Ask James Hardie™

jhcs@jameshardie.com.ph

1 INTRODUCTION

HardieFlex® Wet Area Walls are tough, flexible building products, unsurpassed for economy, versatility and ease of working. They are ideal for use in lightweight construction of timber and light gauge steel framed internal wet area walls.

HardieFlex® Fiber Cement Boards come in a natural grey colour with an off-the-machine smooth surface which will readily accept most site applied finishes.

HardieFlex® Wet Area Walls are perfect for walls in bathrooms, restrooms, kitchens, washing and laundry areas.

The features and benefits of HardieFlex® Wet Area Walls are:

- Durable and impact resistant for longer-lasting bathrooms and kitchens.
- Can be tiled or painted.
- Resistant to damage caused by fire, moisture and termites that will reduce maintenance cost, when installed and maintained correctly.

The specifier or other responsible party for the project must ensure the information and details in this guide are appropriate for the intended application and that specific design and detailing is undertaken for areas which fall outside the scope of this documentation.

IMPORTANT NOTES

- 1 Failure to install, finish or maintain this product in accordance with applicable building codes, regulations, standards and James Hardie's written installation recommendations may lead to personal injury, affect system performance, violate local building codes, and void James Hardie's product warranty.
- 2 All warranties, conditions, liabilities (direct, indirect or consequential) and obligations whether arising in contract, tort or otherwise, other than those specified in James Hardie's product warranty, are excluded to the fullest extent allowed by law. For James Hardie's product warranty information and disclaimers about the information in this manual, see the section at the end of this manual.
- 3 The builder must ensure that the product meets aesthetic requirements before installation. James Hardie will not be responsible for rectifying aesthetic surface variations following installation.
- 4 Make sure your information is up to date. When specifying or installing James Hardie products, ensure you have the current manual. If in doubt, or you need more information, visit www.jameshardie.com.ph or Ask James Hardie™ at +632-895-5427 or 1800-1888-5427.

2 PRODUCT DESCRIPTION

Table 1

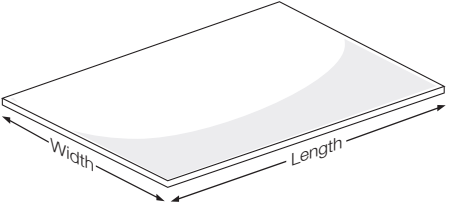
HARDIEFLEX® WET AREA WALLS SHEET SIZES		
		HardieFlex® Wet Area Walls
	Length (mm)	2438mm 2400mm
	Width (mm)	1219mm 1200mm
	Thickness (mm)	6.0mm , 9.0mm , 12.0mm
	Approx. Weight (kg)	28kg for 6mm
All dimensions and masses are approximate and subject to manufacturing tolerance.		

Table 2

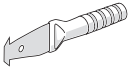
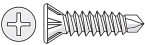
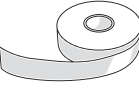
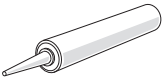
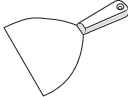
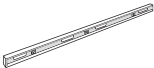
ACCESSORIES / TOOLS SUPPLIED BY JAMES HARDIE			
PRODUCT	DESCRIPTION	PRODUCT	DESCRIPTION
	Scoring Knife Scoring tool for easy cutting of HardieFlex® sheet. Part Code: 80003		HardieFlex® Screw for fixing HardieFlex® sheets to steel frame 0.55 to 1.6m BMT 8 x 32 Screw Part Code: 81040
	HardieFlex® Nails 32 x 2.3mm galvanised nail for fixing HardieFlex® sheet to seasoned timber. Part Code: 81005		HardieFlex® Putty Jointing Compound Appearance: grey paste Pot life at 20°C: 90-120mins. Drying time at 20°C : 60mins Note: Drying time will vary depending on temperature 25kg. Part Code: 82024
			Perforated Paper Tape Joint reinforcing tape used with HardieFlex® sheets. Part Code: 81026

Table 3

ACCESSORIES / TOOLS NOT SUPPLIED BY JAMES HARDIE

James Hardie recommends the following products for use in conjunction with its Wet Area Walls. James Hardie does not supply these products and does not provide a warranty for their use. Please contact the component manufacturer for information on their warranties and further information on their products.

PRODUCT	DESCRIPTION	PRODUCT	DESCRIPTION
	Polyurethane Sealant Suitable sealant for use in movement joints.		Spatula For removal of excess compounds from the panel surface.
	HardieBlade™ Saw Blade 185mm Polydiamond blade, for fast, clean cutting of James Hardie® fiber cement board.		Backing Tape Shiny waxy non-stick tape used at movement joints.
	Level/Straight Edge For checking straightness of underlying flooring.		Notched Trowel For adhesive application to James Hardie substrate.
	Davco K10 Polyurethane Plus/Acrylic Waterproofing Membrane Flexible, single pack water based polyurethane, cures in 6-8 SP. Class III AS4858 compliant; Green Star Specification for V2 IEQ-13 and V1.1 IEQ-11		

GENERAL

HardieFlex® Cement boards are cellulose fiber reinforced cement building products. The basic composition is portland cement, ground sand, cellulose fiber and water. **HardieFlex®** Cement Boards are manufactured to AS/NZS 2908.2 'Cellulose-Cement Products Part 2: Flat Sheets' (ISO 8336 'Fiber Cement Flat Sheets'). **HardieFlex®** Cement Boards are classified Type B, Category 2 in accordance with AS/NZS 2908.2.

Asbestos-Free: **HardieFlex®** Cement Boards do not contain any asbestos (chrysotile) fibers.

DURABILITY

Resistance to Moisture/Rotting

HardieFlex® Cement Boards have demonstrated resistance to permanent moisture-induced deterioration (rotting) by passing the following tests in accordance with AS/NZS 2908.2:

- Water Permeability (Clause 8.2.2).
- Warm Water (Clause 8.2.4).
- Heat Rain (Clause 6.5).
- Soak Dry (Clause 8.2.5).

Resistance to Fire

HardieFlex® Fiber Cement Boards are suitable where non-combustible materials are required in accordance with C1.12 of the Building Code of Australia.

James Hardie building products have been tested by CSIRO in accordance with AS/NZS 3837 and are classified as conforming to Group 1 material (highest and best result possible), with an average specific extinction area far lower than the permissible 250m²/kg, as referenced in Specification C1.10a of the BCA.

Resistance To Termite Attack

Based on testing completed by CSIRO Division of Forest Products and Ensis Australia, James Hardie building products have demonstrated resistance to termite attack.

STORAGE AND HANDLING

Sheets must be installed in a dry state. Before installation, sheets must be laid flat under cover on a smooth level surface clear of the ground to avoid

exposure to water or moisture, etc. **HardieFlex®** Cement Board is resistant to permanent moisture damage when installed as directed (as outlined in this manual).

QUALITY

James Hardie conducts stringent quality checks to ensure that any product manufactured falls within our quality spectrum. It is the responsibility of the builder to ensure that the product meets aesthetic requirements before installation. James Hardie will not be responsible for rectifying obvious aesthetic surface variations following installation.

3 DESIGN / FRAMING

SCOPE

This manual covers the use of the HardieFlex® Wet Area Walls in an internal residential wet area wall application over a seasoned timber frame or a light gauge steel frame.

DESIGN

General

All design and construction must comply with the appropriate requirements of the current National Building Code of the Philippines and other applicable regulations and standards.

Responsibility

The specifier or other party responsible for the project must ensure that the details in this specification are appropriate for the intended application and that additional detailing is performed for specific design or any areas that fall outside the scope of this specification.

FRAMING

General

HardieFlex® Wet Area Walls can be fixed to either timber or light gauge domestic type steel framing. The framing used must comply with the relevant building regulations and standards and the requirements of this manual.

NOTE: Ensure all sheet edges are fully supported with framing members.

Timber

Use only seasoned (kiln dried) timber. Unseasoned timber must not be used, as it is prone to shrinkage and can cause the sheets and frames to move. The minimum stud width at joints is 35mm.

Steel

The size of the steel stud framing should be 0.55mm-1.6mm base metal thickness (BMT).

Steel sections should be galvanized or zincalume. Frame members must not be less than 35mm wide.

Preparation Of Frame

Ensure frame is square and straight, and work from a central datum line. Frames must be straight to provide a flush face to receive the sheeting.

A suggested maximum tolerance of between 3mm and 4mm in any 3m length of frame will give best results.

HardieFlex® Wet Area Walls will not straighten excessively warped or distorted frames and any warping may still be visible after the internal lining is applied.

4 SHEET INSTALLATION

SHEET LAYOUT

Install the HardieFlex® Wet Area Walls in the direction of the framing members i.e. place the long edges of the sheets parallel to the main framing members.

Table 4

	Studs	Noggings
HardieFlex® Wet Area Walls	406 mm	1219 mm

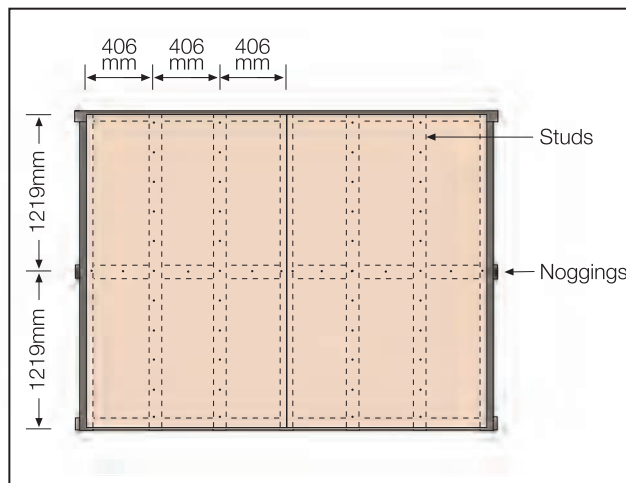


FIGURE 1 SHEET AND FRAME LAYOUT

WINDOW AND DOOR OPENINGS

In normal situations, the end joints should not coincide with corners of doors or window openings. Ensure sheet edges do not coincide with the side of the door or window opening by a minimum distance of 200mm. Refer to Figure 2. Where end joints occur, they should be fully supported by framing and coincide with the centerline of a stud, nogging or plate. This is to ensure sufficient edge clearance for nailing or screwing of adjoining sheets along the sheet edges.

All construction must be carried out in accordance with the relevant building regulations and standards.

Steel studs must be firmly secured to top and bottom plates, and frames must be rigid and not rely on the HardieFlex® Wet Area Walls to provide bracing.

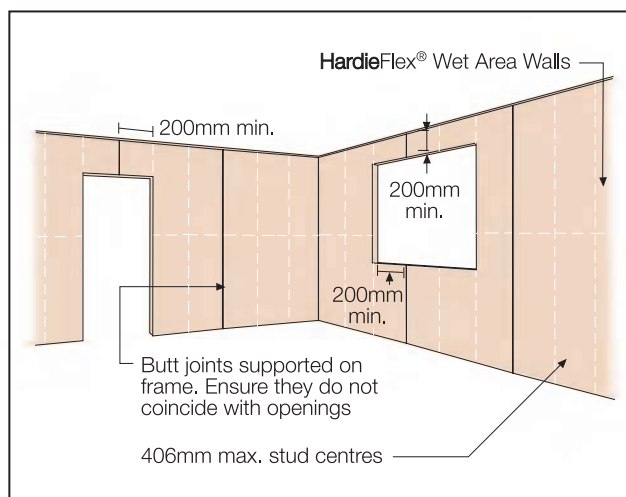


FIGURE 2 SHEET LAYOUT WITH DOOR AND WINDOW OPENING

FASTENERS

For steel frames, use HardieFlex® Screw (8x32mm). For timber frames, use (2.3x32mm) galvanized HardieFlex® Nails.

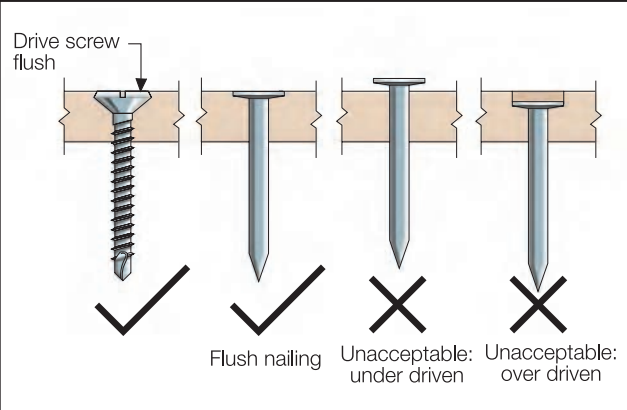


FIGURE 3 FASTENER DEPTH

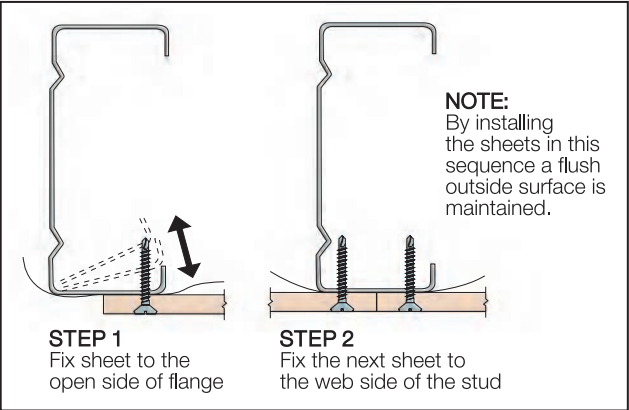


FIGURE 4 SCREW FASTENING

FIXING TO WALLS

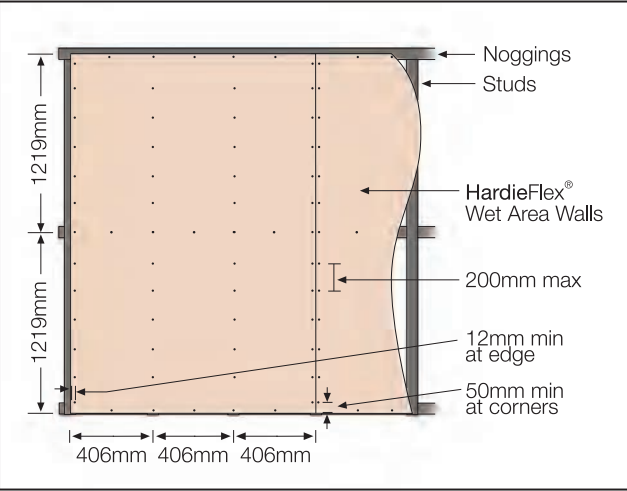


FIGURE 5 FASTENER SPACING

FIXING TO WALLS (first sheet)

Place 6mm off-cut packers along the floor as a temporary support for the sheets. This allows provision for frame movement. Care should be taken to ensure that this gap does not become filled with cement, adhesive or grout during floor tiling procedures. See Figure 6.

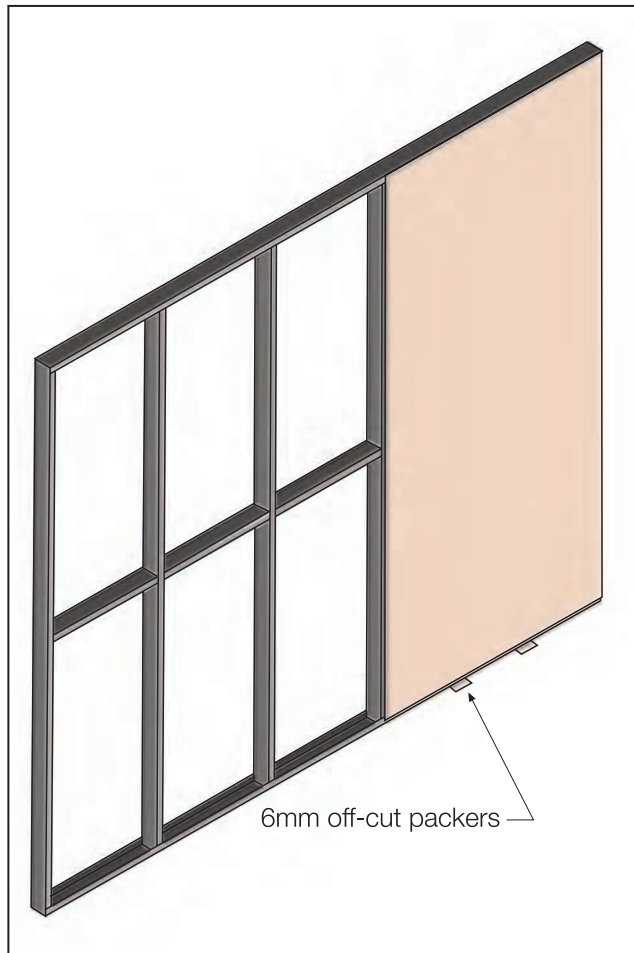


FIGURE 6 FIRST SHEET INSTALLATION

FIX REMAINING SHEETS

Fix the next HardieFlex® Wet Area Wall. Ensure that there is no space between HardieFlex® Wet Area Walls. See Figure 7.

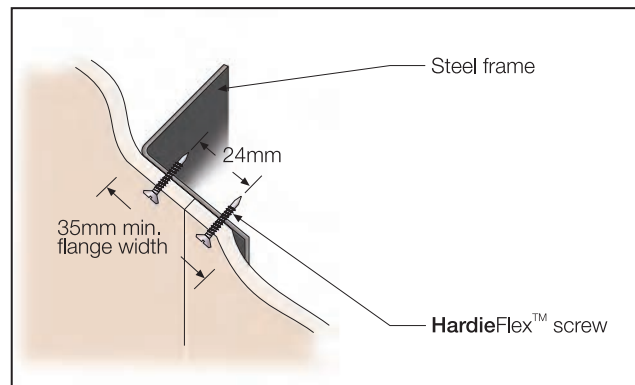
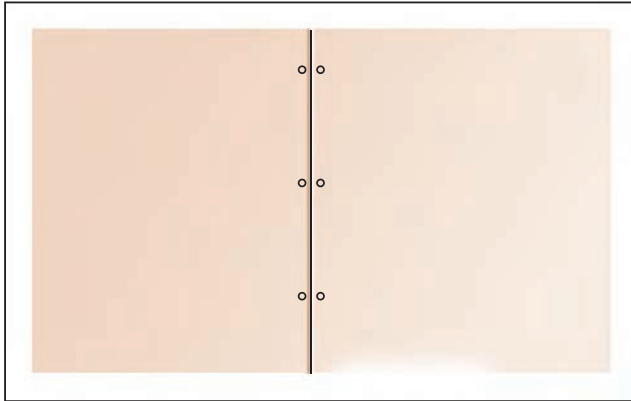


FIGURE 7 THERE SHOULD BE NO GAPS BETWEEN SHEETS

Fix sheets starting from the center of sheet working outwards to avoid any drumminess. Ensure that the sheet is held hard against the frame during nailing or screwing to minimize fastener breakout in the back of the sheet.

5 HIGH PERFORMING JOINTS (H.P.J)

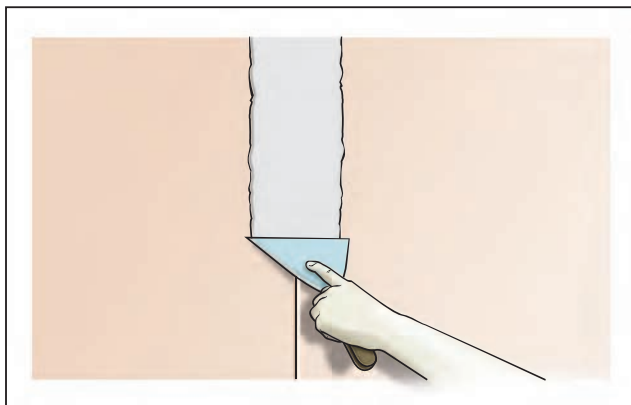
JOINTING BETWEEN HARDIEFLEX® WET AREA WALLS



Step 1 Install HardieFlex® Wet Area Walls without gap, as shown in Figure 7.



Step 2 Wet the board with clean water. Use clean sponge and water and ensure board is clean of dust.



Step 3 Apply 1st Coat of HardieFlex® Putty Jointing Compound.
Apply enough to cover the joint at least 150mm wide. Use a clean spatula. Apply maximum of 1 meter to ensure compound remains pliable.

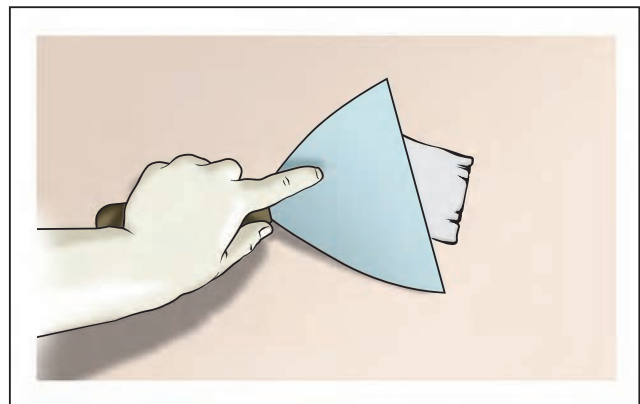


Step 4 Immediately install Perforated Paper Tape
Ensure there are no voids under tape using the spatula and remove excess compound. The paper tape should be slightly moistened with a mist spray gun.



Step 5 Immediately apply 2nd Coat of HardieFlex® Putty Jointing Compound.
Make sure to cover the paper tape. The ideal coat spread is 250mm to even out the surface of the board. Apply another coat if necessary. Let dry and then sand.

FASTENER HEADS



COVER ALL FASTENER HEADS WITH THE HARDIEFLEX® PUTTY COMPOUND

JOINTING BETWEEN HARDIEFLEX® WET AREA WALLS TO CONCRETE WALL OR CEILING OR FLOOR

Install HardieFlex® Wet Area Walls with a 1:1 sealant filled gap. Insert a backing tape in between the gap to create 1:1 ratio.

Width = Depth

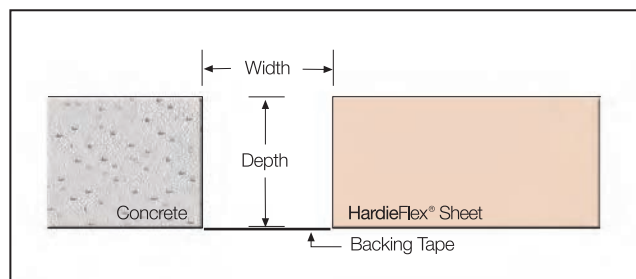


FIGURE 8 1:1 GAP

Apply polyurethane sealant to fill the void flush with sheet.

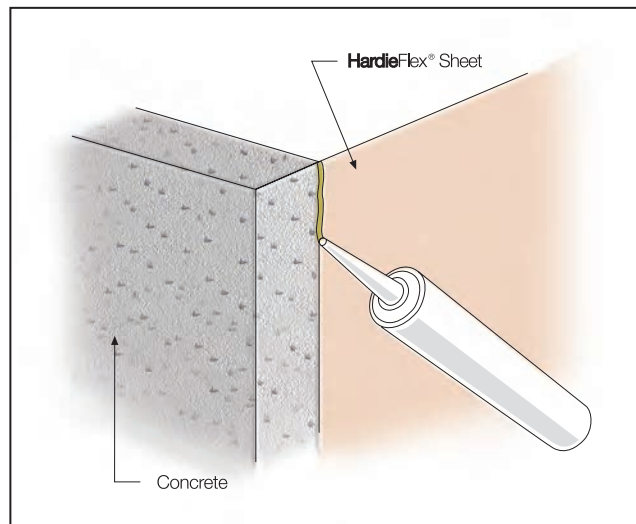
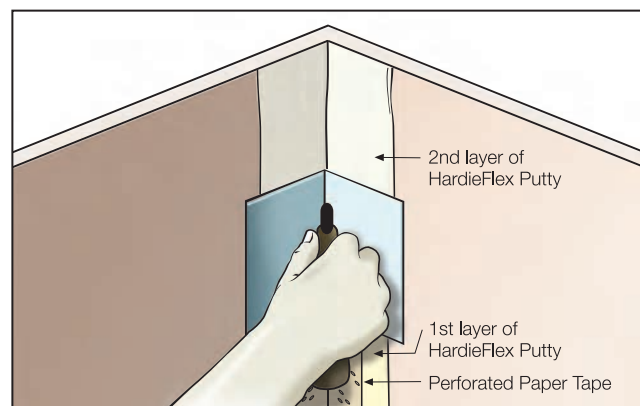


FIGURE 9 POLYURETHANE SEALANT

INTERNAL CORNERS

Setting of internal corners is required for untiled applications only as follows:

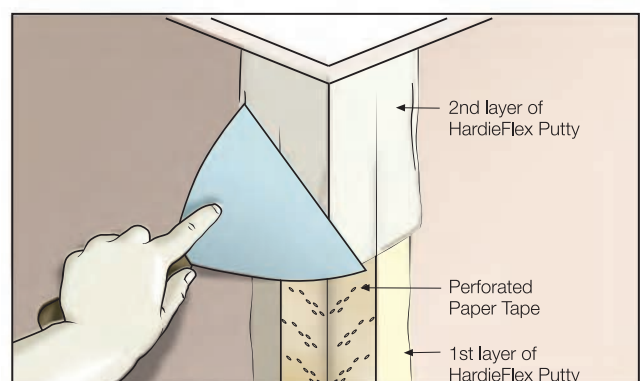
- Wet HardieFlex® Wet Area Walls corners.
- Apply HardieFlex® Putty Compound to both sides of the corner using a 70mm broadknife.
- Fold paper tape to form an angle and embed into the corner.
- Apply another layer of HardieFlex® Putty compound over the angle and smooth with the corner tool.
- Allow to fully dry before sanding.



EXTERNAL CORNERS

Setting of external corners is required for untiled applications only as follows:

- Wet HardieFlex® Wet Area Walls corners.
- Apply HardieFlex® Putty Compound to both sides of the corner using a 70mm broadknife.
- Fold paper tape to form an angle and embed into the corner.
- Apply another layer of HardieFlex® Putty Compound over the angle and smooth with the corner tool.
- Allow to fully dry before sanding.



CONTROL JOINTS

Control Joints are required in long runs of HardieFlex® Wet Area Walls in both directions. These joints are designed to take up the structural movement between the sheets and the building frame.

Table 5

	Steel Framing		Timber Framing
	0.55 - 0.75mm BMT	0.8 - 1.6mm BMT	
Untiled	9.0m	6.0m	7.2m
Tiled	4.8m		4.2m

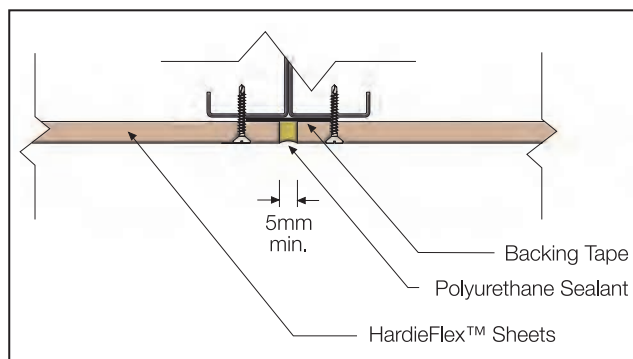


FIGURE 10 VERTICAL CONTROL JOINT

HORIZONTAL CONTROL JOINT

3.6m in height or at the sheet end when using shorter sheets.

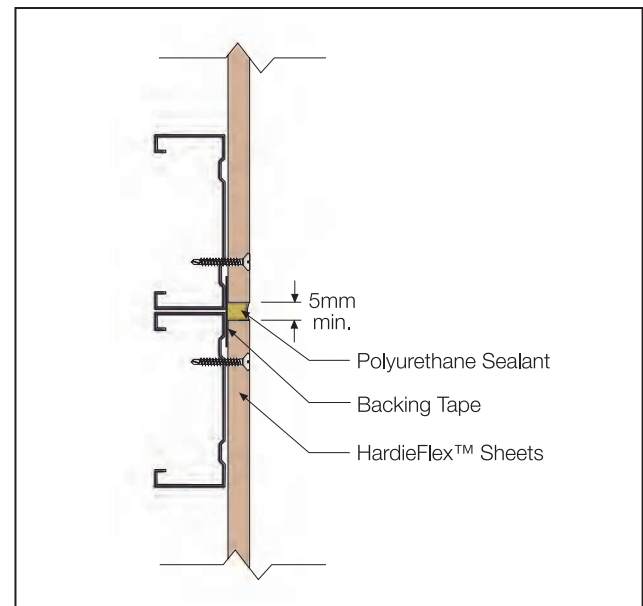


FIGURE 11 HORIZONTAL CONTROL JOINT

6 WALL FLASHING

FLOOR/WALL JUNCTION - OUTSIDE SHOWER AREA

All floor/wall junctions must be sealed with a flashing. The horizontal leg of flashing must be 50mm minimum and the vertical leg must project 40mm minimum above the highest point of the finished floor surface.

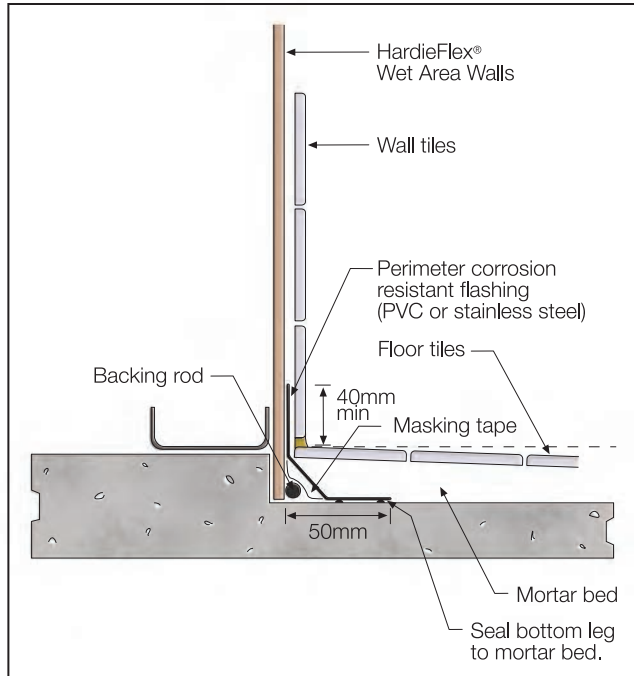


FIGURE 12 FLOOR/WALL JUNCTION FLASHING - OUTSIDE SHOWER AREA

CORNER FLASHING

Vertical corner flashing must be used in shower areas and extended to 1800mm above the finished floor level or base of the bath (where a shower is over the bath).

External vertical corner flashing must have legs of sufficient width to overlap the wall lining by a minimum of 75mm, see Figure 13. The HardieFlex® Wet Area Walls must be sealed at the flashing with a polyurethane sealant.

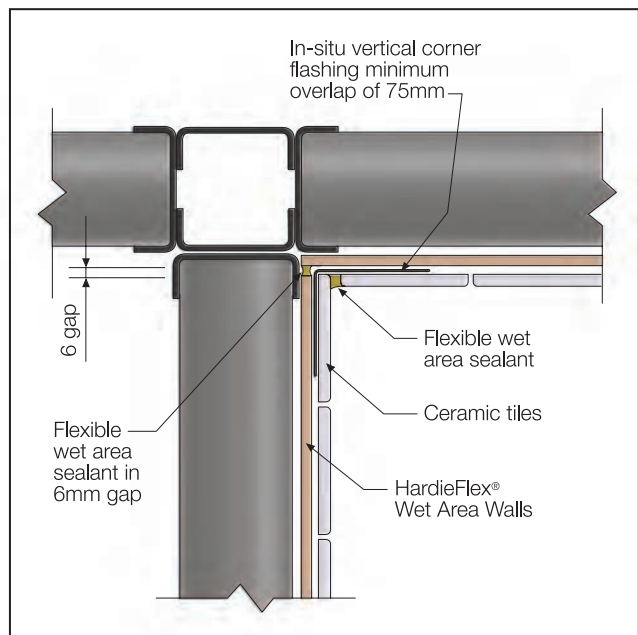


FIGURE 13 VERTICAL CORNER FLASHING

7 WATERPROOFING

For wet areas, the waterproofing requirements of all relevant codes, standards and regulations must be met. Refer to the manufacturer's specifications on application, compatibility and suitability of waterproofing membrane.

James Hardie recommends use of acrylic waterproofing membrane. Refer to the manufacturer's specifications on application.

ENCLOSED SHOWERS

For enclosed showers, the entire floor area must be waterproofed. The waterproofing membrane must extend 150mm up the wall or 25mm above the maximum possible water level in the shower compartment, whichever is higher.

If tiling the shower area you have two waterproofing options. If your shower area, is untiled you have only one option.

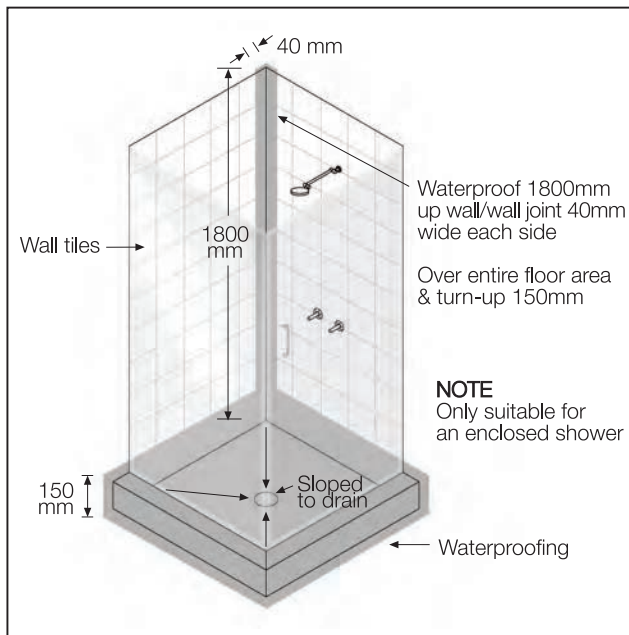


FIGURE 14 ENCLOSED SHOWER, TILED WALLS

NOTE

Enclosed shower fully contains water within the shower area only.

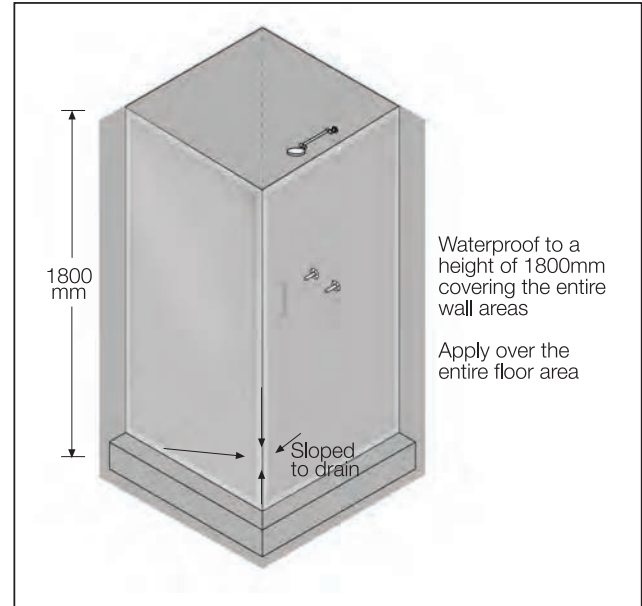


FIGURE 15 ENCLOSED SHOWER, TILED OR UNTILED WALLS

UNENCLOSED SHOWERS

Where the shower is unenclosed, you must waterproof the flooring for a distance extending 1800mm from the shower rose, projecting onto the floor. Tiles must also extend 1800mm from the shower rose.

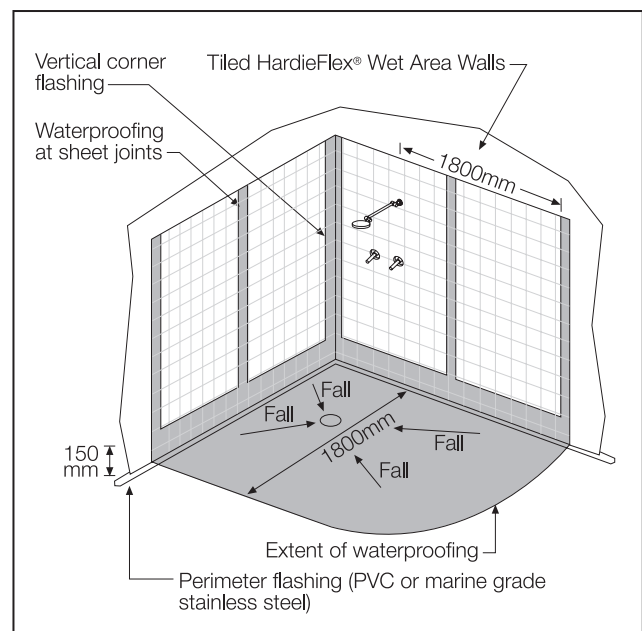


FIGURE 16 UNENCLOSED, TILED WALLS

8 TILING

TILING

Apply tile adhesive as per manufacturer's specification.
Install the tiles.

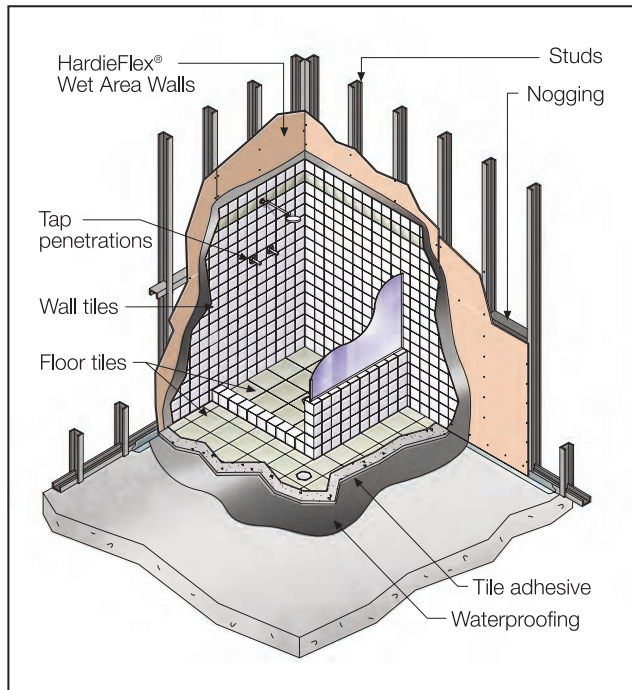


FIGURE 17 TYPICAL BATHROOM FINISH

TILED FINISHES

The thickness of tiles used is restricted based on the thickness of the sheet as shown in Table 6.

Table 6

MAXIMUM TILE THICKNESS	
HARDIEFLEX® WET AREA WALLS (MM)	MAXIMUM TILE THICKNESS (MM)
6	13
9	18
12	>25*

*Support angles are recommended

9 BATHS, BASINS AND TUBS

BATHS

Baths and spas must be sufficiently bedded to prevent cracking and must be recessed into the wall framing to allow the finished wall lining to pass down over the perimeter rim rebate. For the extent of sealing of junctions to a bath enclosure without a shower, see Figure 18.

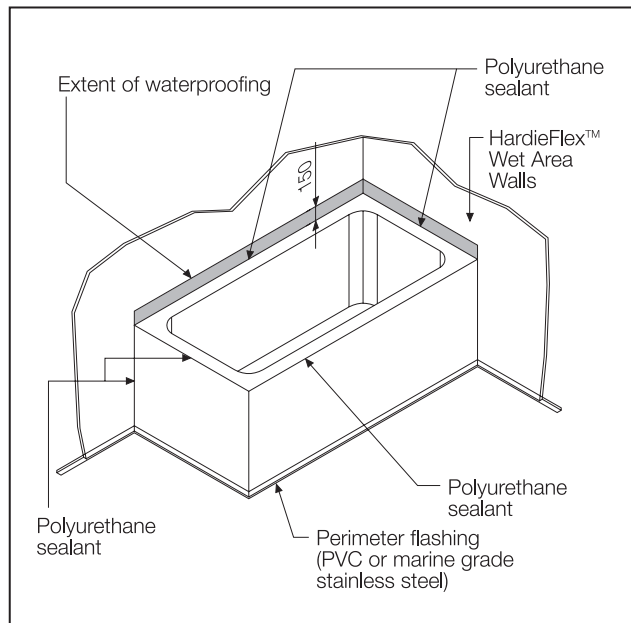


FIGURE 18 BATH WITHOUT SHOWER

BASINS AND TUBS

You must seal vessels such as hand basins and tubs that abut walls, see Figure 19.

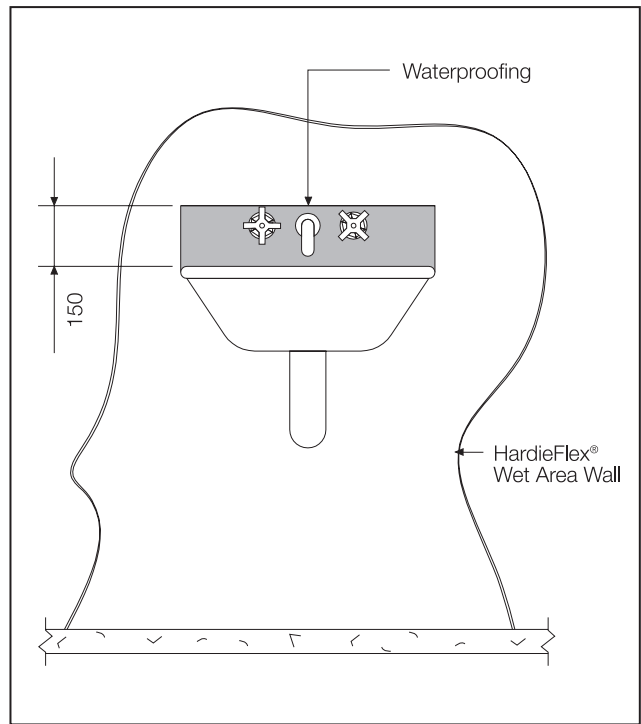


FIGURE 19 WATERPROOFING VESSELS ABUTTING WALLS

10 PENETRATIONS

WALL PENETRATIONS

Use a hole saw to form a neat cut for plumbing fixtures in **HardieFlex® Wet Area Walls**. Taps should be adequately sealed to maintain the waterproof integrity of the wall, see Figures 20 and 21.

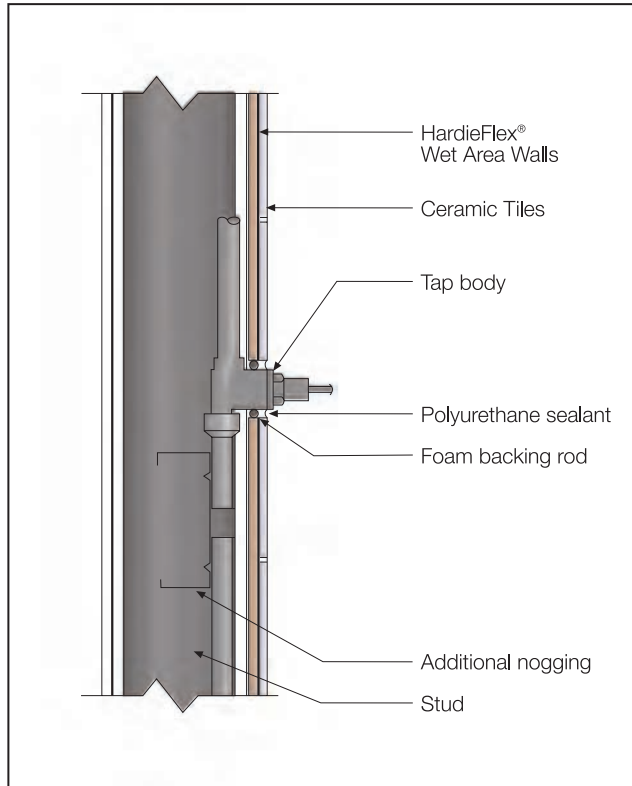


FIGURE 20 TYPICAL PLUMBING PENETRATION - SECTION

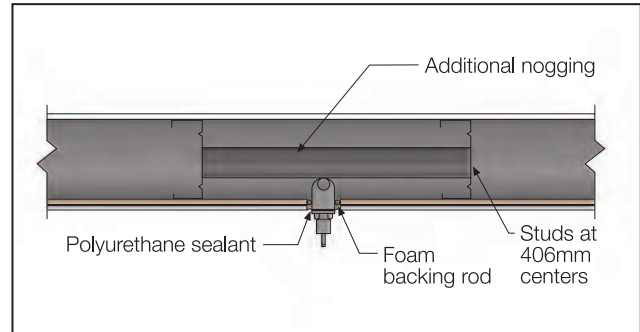


FIGURE 21 TYPICAL PLUMBING PENETRATION - PLAN

11 SAFE WORKING PRACTICES

HardieFlex® Fiber Cement Board has No Asbestos and No Chrysotile fibers.

WARNING: DO NOT BREATHE THE DUST AND CUT ONLY IN WELL VENTILATED AREA

James Hardie products contain sand, a source of respirable crystalline silica which is considered by some international authorities to be a cause of cancer from some occupational sources. Breathing excessive amounts of respirable silica dust can also cause a disabling and potentially fatal lung disease called silicosis, and has been linked with other diseases. Some studies suggest smoking may increase these risks. During installation or handling: (1) work in outdoor areas with ample ventilation; (2) minimize dust when cutting by using either a score and snap knife, fiber cement shears or, where not feasible, use a HardieBlade™ Saw Blade and dust-reducing circular saw attached to a HEPA vacuum; (3) warn others in the immediate area to avoid breathing dust; (4) wear a properly-fitted, approved dust mask or respirator (e.g. N95) in accordance with applicable government regulations and manufacturer instructions to further limit respirable silica exposures. During cleanup, use HEPA vacuums or wet cleanup methods – never dry sweep. For further information, refer to our installation instructions and Material Safety Data Sheets available at www.jameshardie.com.ph. FAILURE TO ADHERE TO OUR WARNINGS, MATERIAL SAFETY DATA SHEETS, AND INSTALLATION INSTRUCTIONS MAY LEAD TO SERIOUS PERSONAL INJURY OR DEATH.

JAMES HARDIE RECOMMENDED SAFE WORKING PRACTICES

CUTTING OUTDOORS

1. Position cutting station so that wind will blow dust away from the user or others in working area.
2. Use one of the following methods based on the required cutting rate:
 - Best**
 - Score and snap
 - Hand guillotine
 - Fibreshear
 - Better**
 - Dust reducing circular saw equipped with HardieBlade™ Saw Blade and HEPA vacuum extraction.
 - Good**
 - Dust reducing circular saw equipped with HardieBlade™ Saw Blade.

CUTTING INDOORS

- Cut only using score and snap, hand guillotine or fibreshears (manual, electric or pneumatic).
- Position cutting station in a well-ventilated area.

SANDING/REBATING/DRILLING/OTHER MACHINING

When sanding/rebating/drilling or machining you should always wear a N95 dust mask and warn others in the immediate area.

IMPORTANT NOTES

1. For maximum protection (lowest respirable dust production), James Hardie recommends always using "Best" - level cutting methods where feasible.
2. NEVER use a power saw indoors.
3. NEVER use a circular saw blade that does not carry the HardieBlade™ logo.
4. NEVER dry sweep - Use wet suppression or HEPA vacuum.
5. NEVER use grinders.
6. ALWAYS follow tool manufacturers' safety recommendations.

P1 or P2 respirators should be used in conjunction with above cutting practices to further reduce dust exposures. Additional exposure information is available at www.jameshardie.com.ph to help you determine the most appropriate cutting method for your job requirements. If concern still exists about exposure levels or you do not comply with the above practices, you should always consult a qualified industrial hygienist or contact James Hardie for further information.

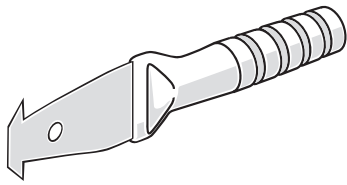
CUTTING

Refer to Recommended Safe Working Practices before starting any cutting or machining of product.

Score and Snap

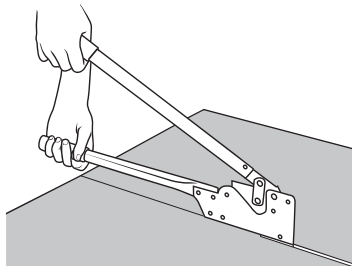
'Score-and-Snap' is a fast and efficient method of cutting **HardieFlex®** Cement Boards using James Hardie's special tungsten tipped score-and snap knife.

- Preferably score from the face side of the **HardieFlex®** Cement Board.
- Score against a straight edge and repeat the action to obtain adequate depth for clean break - normally 1/3 of sheet thickness.
- Snap upwards to achieve break.
- Smooth any rough edges with a rasp or sandpaper.



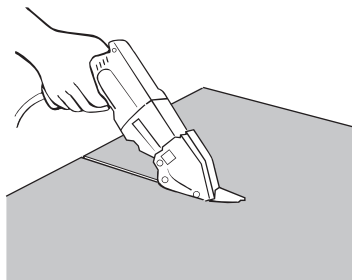
Hand Guillotine

Make guillotine cut on the off-cut side of line to allow for the thickness of the blade.



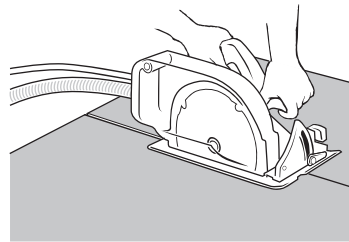
Fibreshear Heavy Duty

An electrically powered, fast, clean and effortless way of cutting James Hardie building products, especially around curves such as archways. Make fibreshear cut on the 'off-cut' side of the line to allow for the thickness of the shear.



HardieBlade™ Saw Blade

The **HardieBlade™** Saw Blade used with a dust-reducing saw and HEPA vacuum extraction allows for fast, clean cutting of James Hardie fibre cement products. A dust-reducing saw uses a dust deflector or a dust collector which can be connected to a vacuum system. When sawing, clamp a straight-edge to the sheet as a guide and run the saw base plate along the straight edge when making the cut.



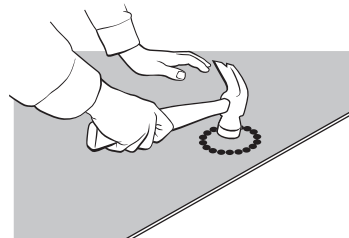
HOLE-FORMING

For smooth clean cut circular holes:

- Mark the center of the hole on the sheet.
- Pre-drill a pilot hole.
- Using the pilot hole as a guide, cut the hole to the appropriate diameter with a hole saw fitted to a heavy duty electric drill.

For irregular holes:

- Small rectangular or circular holes can be cut by drilling a series of small holes around the perimeter of the hole then tapping out the waste piece from the sheet face.
- Tap carefully to avoid damage to sheets, ensuring the sheet edges are properly supported.



12 WARRANTY

James Hardie Philippines, Inc. ("James Hardie") warrants to the first purchaser of its fiber cement flat sheet products known as **HardieFlex®** Sheet (the "Product") for a period of 10 years from the date of purchase, that: the product will be free from defects due to defective factory workmanship or materials and, subject to compliance with the conditions below, will be resistant to cracking, rotting, fire and damage from termite attacks to the extent set out in James Hardie's relevant published literature current at the time of installation.

The recommendations in James Hardie's brochures are formulated along the lines of good building practice but are not intended to be an exhaustive statement of all relevant data. As the successful installation of these systems depends on numerous factors outside the control of James Hardie, such as quality of workmanship and particular design requirements, James Hardie accepts no responsibility for or in connection with the quality of the recommendations or the systems or their suitability for any purpose when installed.

CONDITIONS OF WARRANTY

This warranty is subject to the following conditions:

- The customer must provide proof of purchase and make a claim within 30 days after the defect would have become reasonably apparent.
- The Product must be stored and installed and maintained strictly in accordance with the relevant James Hardie literature current at the time of installation and in accordance with good building practice. The **HardieFlex®** cement board must only be used for internal walls.
- The Product must be maintained in accordance with the relevant James Hardie literature, and no liability or responsibility whatsoever will be accepted in relation to any Product which has not been properly maintained.
- James Hardie will not be liable for claims, damages or defects arising from, or in any way attributable to: poor workmanship; settlement or structural movement and/or movement in materials to which the Product is attached; incorrect design

of the structure; misuse, such as chicken houses, pigpens, cattle barns, farms, hydroponics vegetable houses, hot houses, saunas, buried applications, furnaces, heat-rooms or kilns; acts of God including earthquakes, cyclones, floods or other severe weather conditions or unusual climatic conditions; efflorescence or performance of paint or coatings, jointing compounds, fasteners and other accessories applied to the Product or lack thereof; growth of mold, mildew, fungi, bacteria, or any organism on any Product surface or Product (whether on the exposed or unexposed surfaces); or, defects due to exposure to corrosive or toxic substances including organic acids.

- James Hardie will not be liable for any other claims, damages or losses (whether direct or indirect) including property damage, personal injury, "consequential loss", economic loss or loss of profits arising in contract or negligence or howsoever arising.
- This warranty is not transferable.

WARRANTY CLAIMS

James Hardie will use its reasonable endeavours to satisfy its obligations under this warranty within 60 days after a claim is made. If James Hardie is reasonably satisfied a claim may be made under this warranty it will either (at its option) supply replacement Products, rectify the Product, or pay for the cost of replacement products or rectification of the affected Product. If meeting a claim under this warranty involves re-coating of sheets, there may be slight colour differences between the original and re-coated sheets due to the effects of weathering and variations in materials over time. After the 1st year through to the 10th year, James Hardie may reduce any payment it elects to make for defective Products by 10% of the purchase price of the defective Product for each year since its purchase.

Any refund or material replacement or compensation provided by James Hardie pursuant to this warranty will constitute a settlement and release of all claims made by the customer for damages or other relief. This will be the sole, exclusive remedy of the customer unless the customer has rights at law which cannot be excluded or modified, in which case those rights will apply to the customer.

The statements in this warranty constitute the only warranty extended by James Hardie for the Products. James Hardie disclaims all other warranties, expressed or implied, including any implied warranties of merchantability or fitness for a particular purpose, except where the product purchase is subject to the Consumer Act of the Philippines, in which instance the duration of any applicable implied warranty is limited to 10 years or the period provided for by law, whichever is shorter. Nothing in this warranty is to be construed as excluding or modifying any rights of the customer which cannot be excluded or modified at law.

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For more information on **HardieFlex®** Cement Boards or any other James Hardie product call the James Hardie Customer Service Hotline on +632-8955427 or 1800 1888 5427

Ask JamesHardie™

Call 1800 1888 5427

or phone 895 5427

www.jameshardie.com.ph